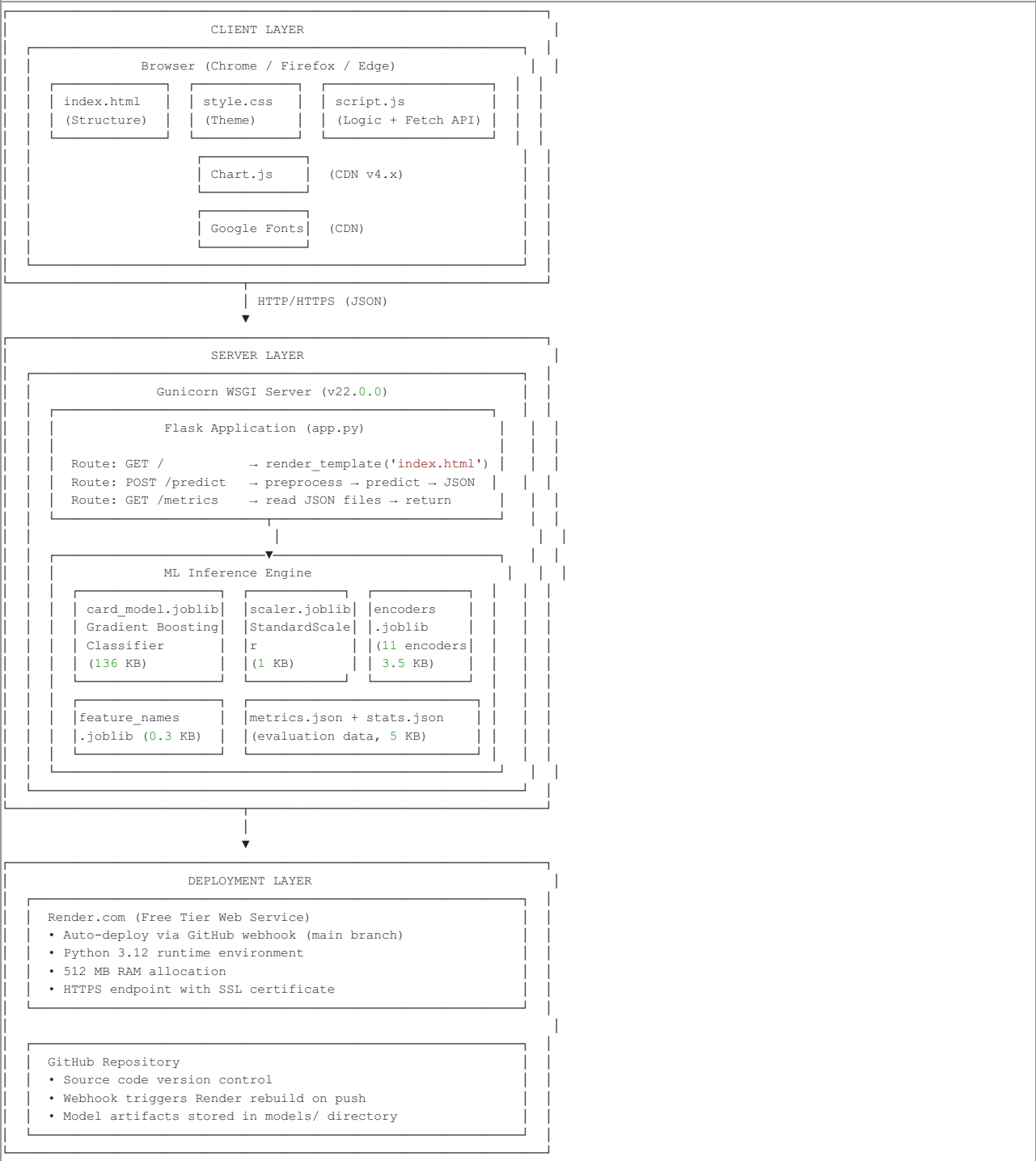


Solution Architecture

CrediShield AI — Credit Card Approval Prediction System

1. High-Level Architecture Diagram



2. Component Interaction Matrix

From	To	Protocol/Method	Data Exchanged	Frequency
Browser	Flask (/)	HTTP GET	HTML page + CSS + JS	On page load
Browser	Flask (/predict)	HTTP POST (JSON)	14 applicant fields → {approved, confidence, risk_level}	Per prediction
Browser	Flask (/metrics)	HTTP GET	→ {metrics: {...}, stats: {...}}	On tab switch
Browser	Chart.js CDN	HTTPS GET	JavaScript library (245 KB)	On first page load
Browser	Google Fonts	CDN HTTPS GET	Font files (Plus Jakarta Sans)	On first page load
Flask	ML Engine	Python function call	18-element numpy array → prediction + probability	Per prediction

Flask	JSON files	File I/O read	metrics.json + stats.json contents	Per /metrics call
GitHub	Render	Webhook (HTTPS)	Push notification → trigger rebuild	Per git push
Render	GitHub	Git clone (HTTPS)	Repository source code + model files	Per deployment

3. Layer Descriptions

3.1 Client Layer (Presentation)

Component	Technology	Size	Purpose
index.html	HTML5 + Jinja2	~15 KB	Page structure: navbar, form, results, tabs, charts
style.css	CSS3	~12 KB	Glassmorphism theme, animations, responsive layouts
script.js	ES6 JavaScript	~14 KB	Form handling, fetch API, SVG animation, localStorage, PDF generation
Chart.js	CDN (v4.x)	245 KB (external)	Interactive bar, doughnut, and pie chart rendering
Plus Jakarta Sans	Google Fonts CDN	~50 KB (external)	Premium typography

3.2 Server Layer (Business Logic)

Component	Technology	Purpose
Gunicorn	v22.0.0	Production WSGI HTTP server; handles concurrent requests
Flask	v3.0.3	Route handling, request parsing, response formatting
Prediction Pipeline	Python	Input validation → feature engineering → encoding → scaling → inference
Error Handling	try-except blocks	Graceful JSON error responses for all failure modes

3.3 ML Inference Engine (Data/Model Layer)

Artifact	Format	Size	Content
card_model.joblib	Binary (Pickle)	136 KB	Fitted GradientBoostingClassifier
scaler.joblib	Binary (Pickle)	1 KB	Fitted StandardScaler (mean + variance for 6 features)
label_encoders.joblib	Binary (Pickle)	3.5 KB	Dictionary of 11 fitted LabelEncoder objects
feature_names.joblib	Binary (Pickle)	0.3 KB	Ordered list: 18 feature column names
metrics.json	JSON	3.8 KB	All 6 models' Accuracy, Precision, Recall, F1, ROC AUC
stats.json	JSON	1.1 KB	Class counts, income categories, age categories

3.4 Deployment Layer (Infrastructure)

Service	Role	Configuration
Render.com	Cloud hosting	Free Tier; auto-deploy from GitHub; Python 3.12
GitHub	Source control	Repository: dineshTech-creator/Credit-Card-Approval-Prediction
Procfile	Process definition	web: gunicorn app:app
requirements.txt	Dependency management	7 pinned production packages
.python-version	Runtime version	3.12

4. Design Patterns Used

Pattern	Where Applied	Benefit
MVC (Model-View-Controller)	Entire application	Separation of concerns: UI, logic, and data are independent
Pre-trained Inference	ML Engine	No training at runtime; fast startup; deterministic predictions
Self-Healing Startup	app.py lines 15–18	Auto-triggers training if model files missing
Configuration as Code	Procfile, vercel.json, .python-version	Infrastructure defined in version-controlled files
Client-Side Caching	localStorage (history)	Reduces server calls; instant history retrieval
Progressive Enhancement	SVG gauge + Chart.js	Core prediction works even if animations fail
API-First Design	/predict and /metrics endpoints	Frontend and backend can evolve independently

5. Security Considerations

Aspect	Implementation
Input Validation	All 14 required fields checked before processing
Error Sanitization	API error messages don't expose stack traces or internal paths
No Database	No SQL injection surface (localStorage is client-side only)
HTTPS	Render provides free SSL certificates for all deployed apps
No Authentication	Current version is a demo — future versions should add login
CORS	Not explicitly configured (same-origin requests only)

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